TELECOM Paris

INSTITUT POLYTECHNIQUE DE PARIS

AI regulation

September 2025



Thomas Le Goff

Ass. Prof. of Law & Technology

What are the main legal risks posed by LLMs ?

General discussion: What are the main legal risks posed by LLMs ?

- **Data protection rules**:
- Complying with the GDPR (having a valid legal basis and respecting all 6 GDPR's principles including transparency, fairness, proportionality…) ;
- Pseudonymization vs anonymization ;
- Technical solutions to make sure that your LLMs do not output personal data

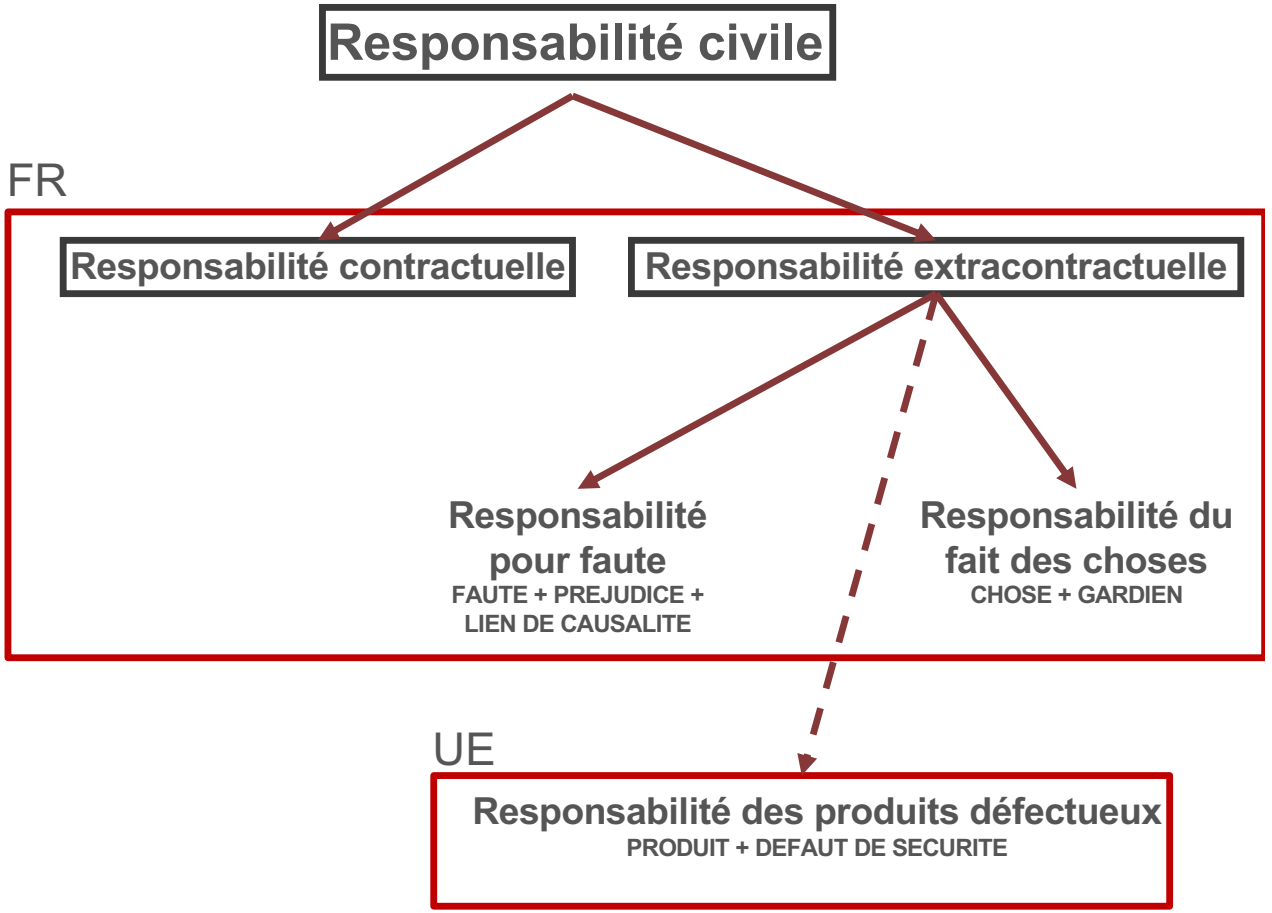
- **Intellectual property**:
- Risks of copyright infringement and how to prevent it (cleaning your training dataset, respect authors' opt-out right, filter your output so that you cannot generate infringing content, secure the compliant use of your LLMs by drafting carefully your contractual terms of use)

- **Liability**:
- Contractual liability (if you breach a contract, remember Air Canada case),
- Civil liability for compensation (prejudice + fault + causal link),
- Criminal liability (if your LLM nudge someone to harm him/herself for example!)

Les régimes de responsabilité

Responsabilité pénale

CONTRAVENTION, DÉLITS, CRIMES
PRÉVUS PAR LE CODE PÉNAL



GDPR obligations

One legal basis AND respect of principles for ALL processing

Legal basis (art 6 GDPR)

« OR »

- Consent
- Necessary to perform of a contract
- Legal obligation
- Vital interest
- Public interest
- Legitimate interest

Processing principles (art 5 GDPR)

- Lawfulness, fairness and transparency
- Purpose limitation
- Data minimisation and accuracy
- Storage limitation
- Integrity and confidentiality
- **Accountability**

« AND »

*Controller must set in place **procedures, technical measures, policies** to ensure compliance, demonstrate clear chains of responsabilities and **be able to demonstrate (« prove ») compliance to internal or external stakeholders**.*

Copyright basics

Copyright basics

- **What's the purpose of copyright?**
  - Promote the advancement of arts and sciences by encouraging creation, investment and sharing
  - Example: motion picture production

- **What is protected?**

« Original expression, by a human, in a fixed medium »

  - **"Expression"**: An idea or method is not protected by copyright (maybe by patent, but that's another story)
    - An algorithm is not protected, but its "expression" , e.g. source code, is protected.
  - **"Original"** expression: something original from your own brain, not copied from someone else, must reflect creative choices
  - **By a human**: original expression by a machine is not protected
  - **In a fixed medium**: written down or recorded

Copyright basics

- **What formalities?**
  - None - the act of expression in a fixed medium is enough
  - But need to prove!
  - Registration helpful in the US
  - © not required but helps for damages in the US

- **What does “protection” give you?**
  - **Exclusive right** to control use - reproduction, display, distribution
    - by territory
    - by technology
    - by time

Fair use – an exception to copyright

- **Why fair use?**
  - Limit the copyright monopoly to strict minimum necessary
  - Encourage other forms of creativity and expression

- **Four factors of fair use exception**
  - **1. the purpose and character of your use**
    - “transformative”?
    - Has the material you have taken from the original work been transformed by adding new expression or meaning?
    - Was value added to the original by creating new information, new aesthetics, new insights, and understandings?

  - **2. the nature of the copyrighted work**
    - “Because the dissemination of facts or information benefits the public, you have more leeway to copy from factual works such as biographies than you do from fictional works such as plays or novels.”

  - **3. the amount and substantiality of the portion taken.**

  - **4. the effect of the use upon the potential market**.

- **Fair use in Europe : the text and data mining exception in EU law**
  - Research purpose
  - Commercial purpose (opt-out)

Do we need rules on AI?

TELECOM Paris

IP PARIS

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Do we need rules on AI?

TELECOM Paris

IP PARIS

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

How China Is Using "Social Credit Scores" to Reward and Punish Its Citizens
By Charlie Campbell / Chengdu

https://time.com/collection/davos-2019/5502592/china-social-credit-score/

Do we need rules on AI?

TELECOM Paris

IP PARIS

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

How China Is Using "Social Credit Scores" to Reward and Punish Its Citizens
By Charlie Campbell / Chengdu

https://time.com/collection/davos-2019/5502592/china-social-credit-score/

Do we need rules on AI?

The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and "might lead to a dystopian future or something," a backer says.

https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html

TELECOM Paris

IP PARIS

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

How China Is Using "Social Credit Scores" to Reward and Punish Its Citizens
By Charlie Campbell / Chengdu

https://time.com/collection/davos-2019/5502592/china-social-credit-score/

Do we need rules on AI?

How AI could take over elections – and undermine democracy



https://theconversation.com/how-ai-could-take-over-elections-and-undermine-democracy-206051

The Secretive Company That Might End Privacy as We Know It

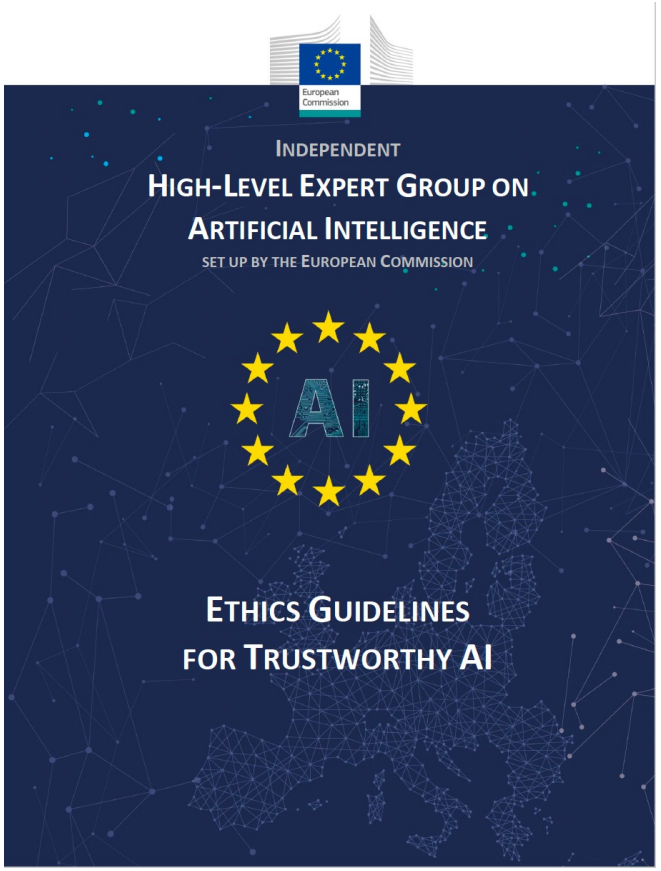
A little-known start-up helps law enforcement match photos of unknown people to their online images — and "might lead to a dystopian future or something," a backer says.

https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html

What is AI ethics?

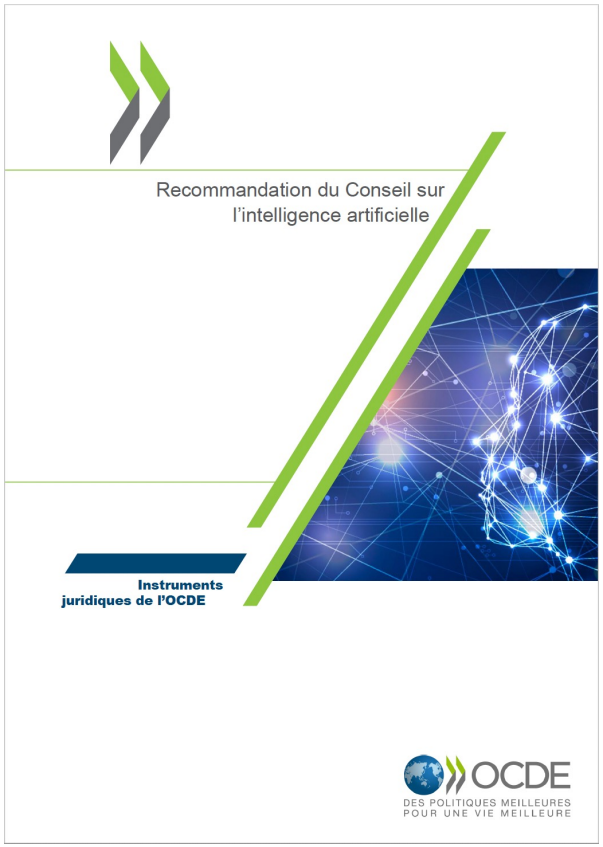
AI ethics principles in Europe

- Human agency and oversight
- Technical robustness and safety
- Privacy and data governance
- Transparency
- Diversity, non-discrimination and fairness
- Societal and environmental well-being
- Accountability



2019

AI ethics principles: OECD countries



Adopted 2019, updated May 2024

- **Inclusive growth, sustainable development and well-being**
- **Respect for the rule of law, human rights and democratic values, including fairness and privacy**
- **Transparency and explainability**
- **Robustness, security and safety**
- **Accountability**
  (risk management by « AI actors » throughout the lifecycle)

AI ethics principles: UNESCO



November 2021

- **Proportionality and Do No Harm**
- **Safety and Security**
- **Right to Privacy and Data Protection**
- **Multi-stakeholder and Adaptive Governance & Collaboration**
- **Responsibility and Accountability**
- **Transparency and Explainability**
- **Human Oversight and Determination**
- **Sustainability**
- **Awareness & Literacy**
- **Fairness and Non-Discrimination**

The multiplication of AI ethics' charters in public organisations and in industry

Perspective | Published: 02 September 2019

The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca & Effy Vayena

Nature Machine Intelligence **1**, 389–399 (2019) | Cite this article

84 AI ethics' charters analysed

- *Global **convergence around five ethical principles** (transparency, justice and fairness, non-maleficence, responsibility and privacy)*
- *Substantive divergence on how these principles are interpreted and how they should be implemented.*

Mélanie Gornet, Simon Delarue, Maria Boritchev, and Tiphaine Viard. 2024. Mapping AI Ethics: a meso-scale analysis of its charters and manifestos. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24), June 3–6, 2024, Rio de Janeiro, Brazil.* ACM, New York, NY, USA, 18 pages. https://doi.org/10.1145/3630106.3658545

436 AI ethics' charters analysed

- *"Topics covered": Societal Impacts ; Company governance ; Policy Recommendations.*
- ***"Missing topics": AI and Work ; AI and Power struggles (injustice, sexism…) ; AI and Ecology.***

What is the legal value of AI ethics' charters?

What is "soft law"?

*« Agreements, principles and declarations that are **not legally binding**. Soft law instruments are predominantly found in the international sphere. Hard law refers generally to legal obligations that are binding on the parties involved and which can be legally enforced before a court. »*

Source: European Center for Constitutional and Human Rights

Three conditions (Conseil d'Etat, rapport annuel, 2013):
- The texts' purpose must be to modify or orient the recipients' behavior, by encouraging their voluntary adherence ;
- Do not create rights nor obligations by itself ;
- Considering their content and the way they were drawn up, they have a degree of formalization and structuring that makes them similar to rules of law.

Shortcomings of soft law

- Not binding => no effect if no voluntary adherence
- Lack of « legal security » and bypass of democratic institutions
- *A way to avoid strict regulation*



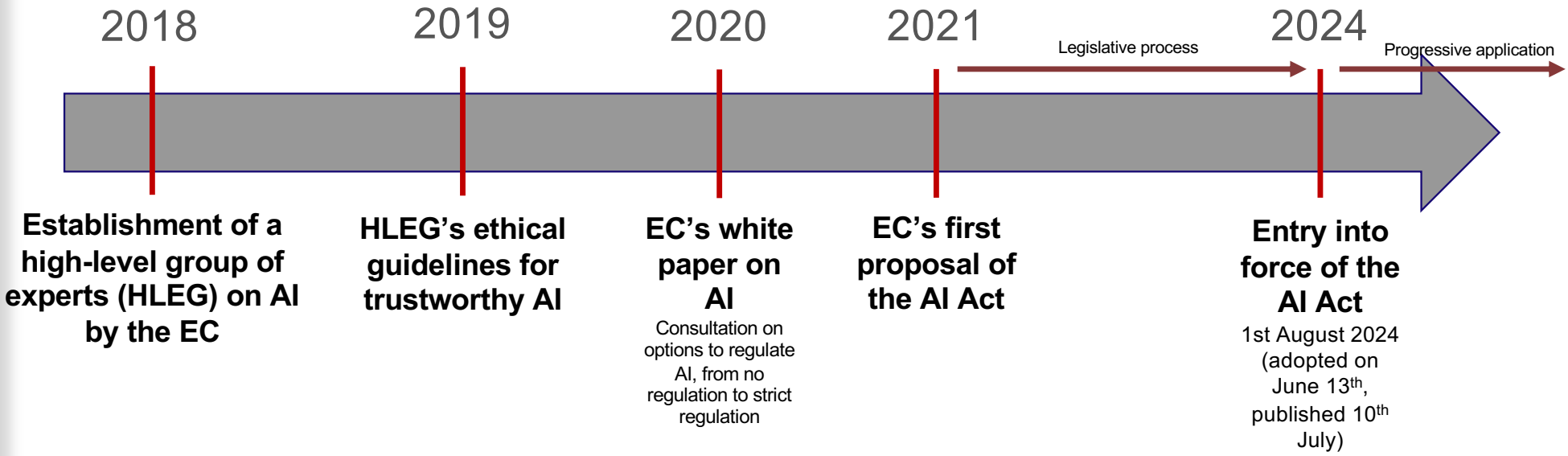
But… "Soft law" can acquire legal force

If traditional institutions grant it legal force (binding standard in an enforceable text, enforced doctrine of authorities, use by a judge to qualify a fault…)

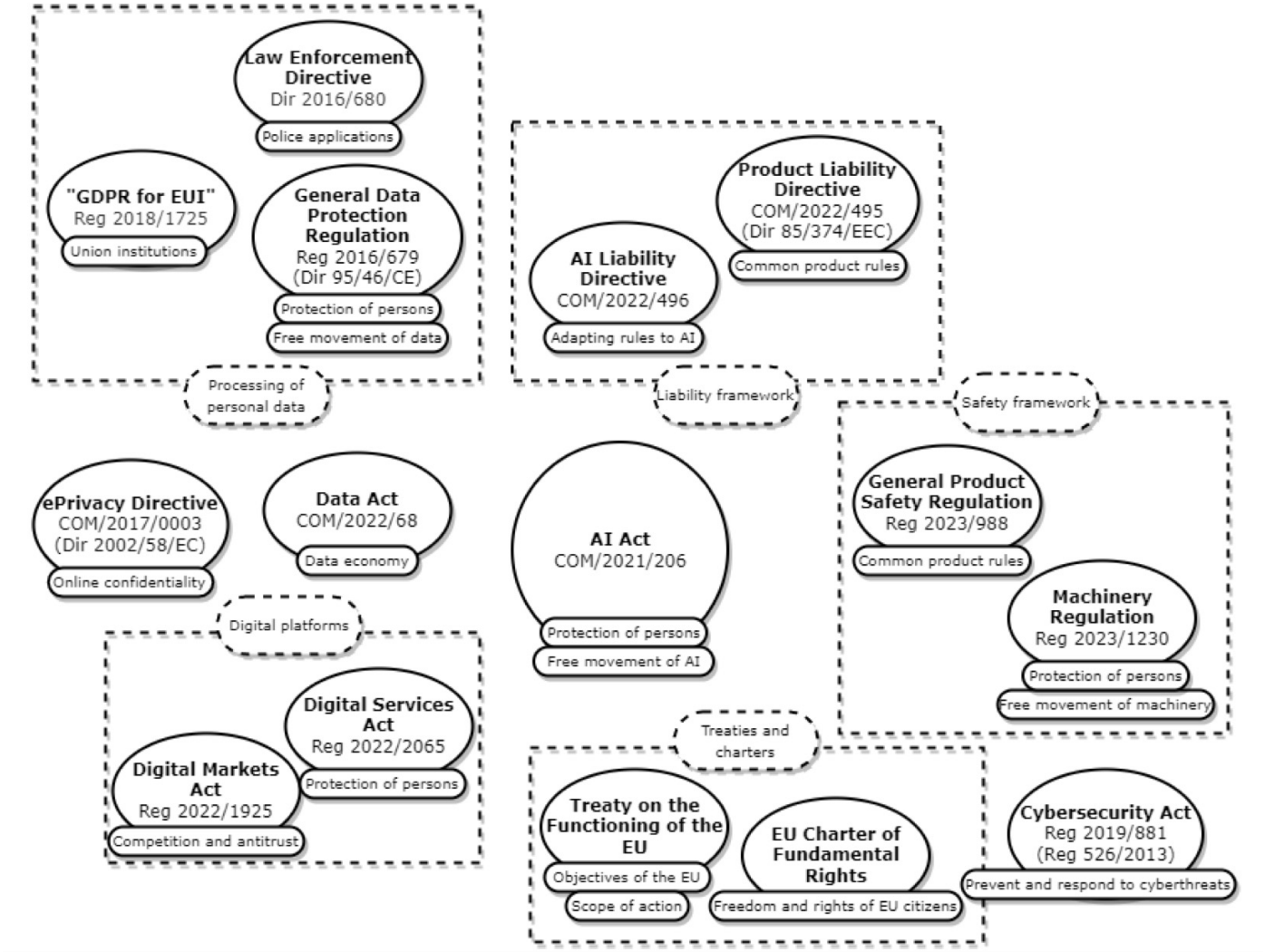
If a company officially adheres to soft law principles (public statement…). Enforcement by authorities (FTC, CNIL…) or courts.

A regulatory response to ethical risks: the European AI Act

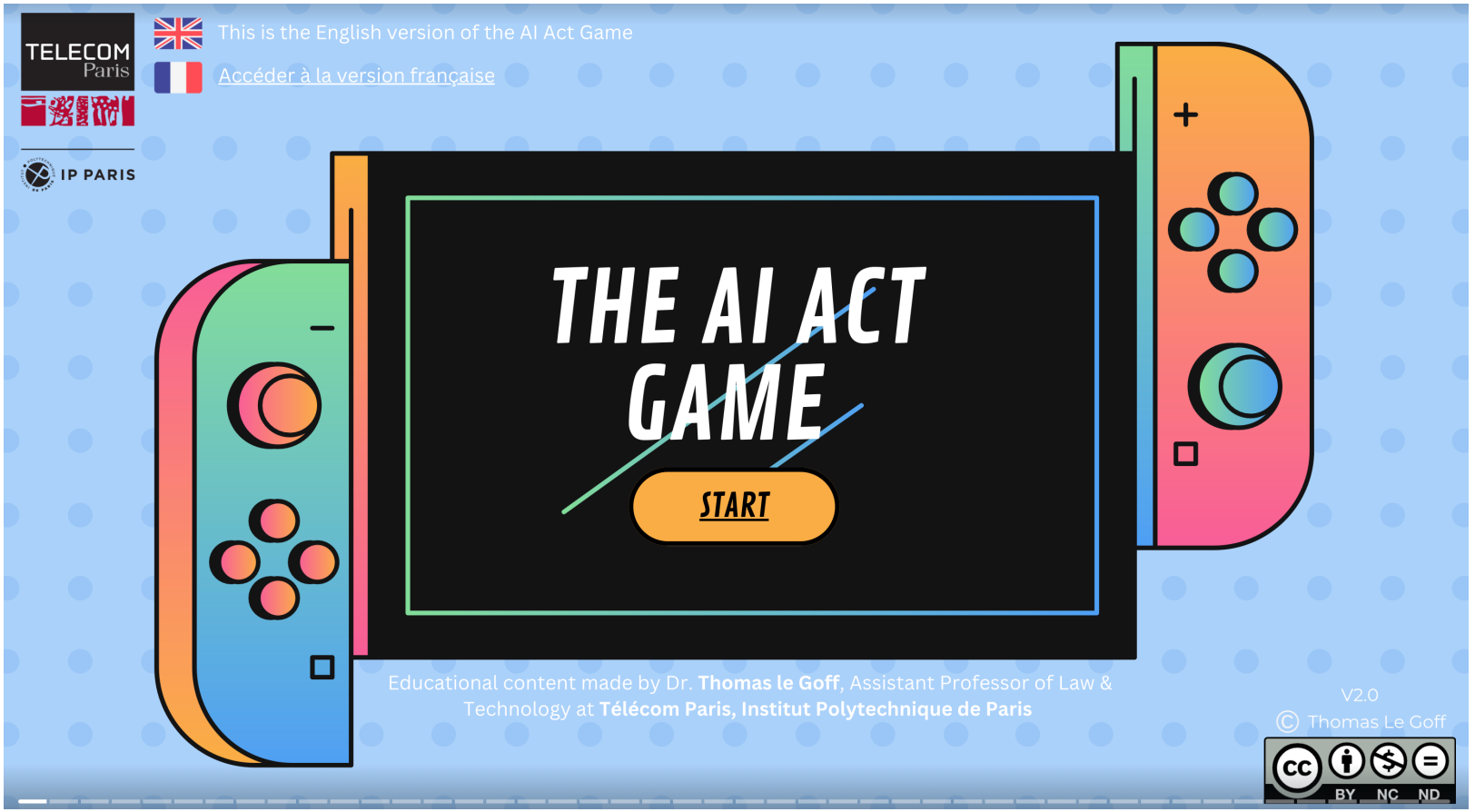
The legislative process



The EU legislative ecosystem of the AI Act



Discover the AI Act



https://www.canva.com/design/DAGF2FfoggE/QpjWW1ghhCr_GMZT_gLJ3A/view?utm_content=DAGF2FfoggE&utm_campaign=designshare&utm_medium=link&utm_source=editor

AI systems' definition in the AI Act

- '***AI* system**' *means* ***a machine-based system*** *designed to* ***operate with varying levels of autonomy****, that may* ***exhibit adaptiveness after deployment*** *and that, for* ***explicit or implicit objectives****,* ***infers, from the input*** *it receives, how to* ***generate outputs*** *such as predictions, content, recommendations, or decisions* ***that can influence physical or virtual environments****.*

AI systems' definition in the AI Act

Optional criteria

- '***AI* system**' *means* ***a machine-based system*** *designed to* ***operate with varying levels of autonomy****, that may* ***exhibit adaptiveness after deployment*** *and that, for* ***explicit or implicit objectives****,* ***infers, from the input*** *it receives, how to* ***generate outputs*** *such as predictions, content, recommendations, or decisions* ***that can influence physical or virtual environments****.*

Optional criteria

Optional criteria

AI systems' definition in the AI Act

- '***AI* system**' *means a machine-based system designed to operate with varying levels of autonomy, that may exhibit adaptiveness after deployment and that,* ***for explicit or implicit objectives, infers, from the input it receives, how to generate outputs*** *such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.*

AI systems' definition in the AI Act

Recital 12

- "The definition should be based on key characteristics of AI systems that distinguish it from simpler traditional software systems or programming approaches and **should not cover systems that are based on the rules defined solely by natural persons to automatically execute operations**"

- "**A key characteristic of AI systems is their capability to infer**. This capability to infer refers to the process of obtaining the outputs (…) and to **a capability of AI systems to derive models or algorithms, or both, from inputs or data**. The techniques that enable inference while building an AI system include machine learning approaches that learn from data how to achieve certain objectives, and logic- and knowledge-based approaches that infer from encoded knowledge or symbolic representation of the task to be solved"

Who should comply with the AI Act ?

- "**Provider**" : body that places an AI system on the market or puts it into service under its own name

→ Responsible for ensuring *ex-ante* compliance with the requirements of the AI Act.

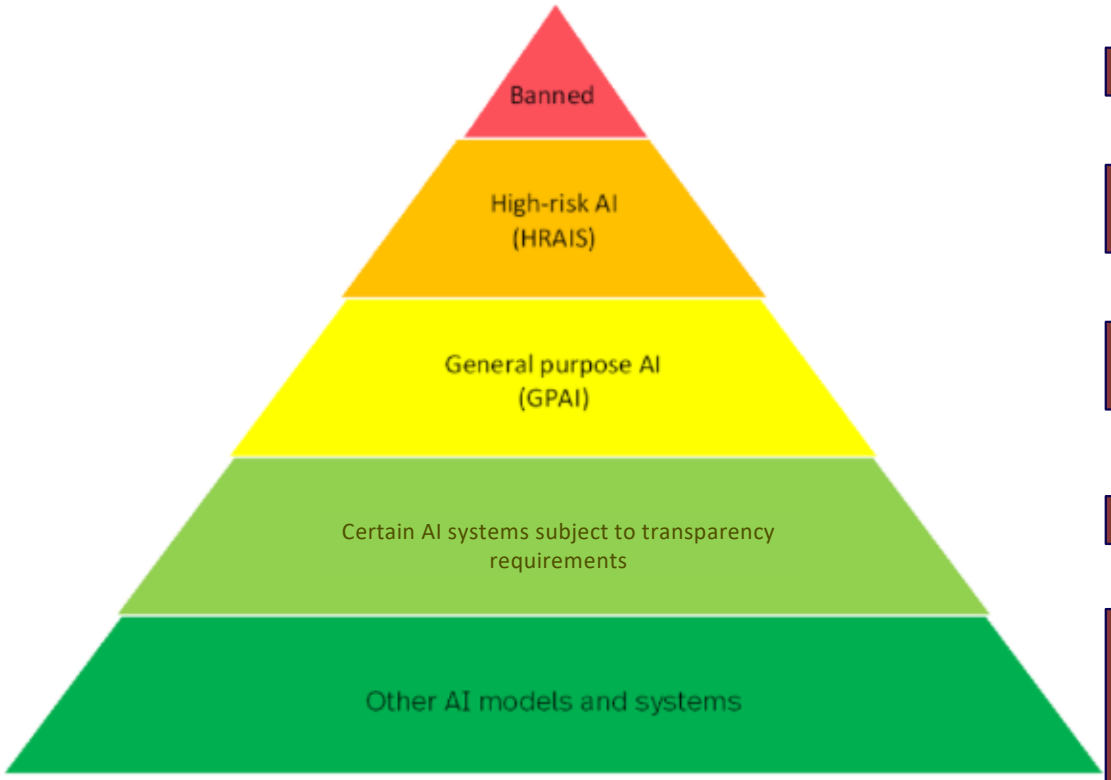
- "**Deployer**" : body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity

→
- Becomes a provider if it places the system on the market or puts it into service under its own name
- Should ensure that the system has undergone the appropriate conformity assessment (CE marking)
- Should comply with providers' instructions of use

- "**Distributor**" (makes an AI system available on the EU market) and "**Importer**" (places an AI system that bears the name of an entity from a third country on the EU market)

→ Should ensure that the system has undergone the appropriate conformity assessment (CE marking)

The AI pyramid of risks



Source : https://www.simmons-simmons.com/en/features/artificial-intelligence/clzb255y300riuxgk8glgiyrv/eu-artificial-intelligence-act

Prohibition

Strict compliance obligations (data governance, risk assessment…)

Specific obligations (transparency, systemic risks assessment…)

Transparency

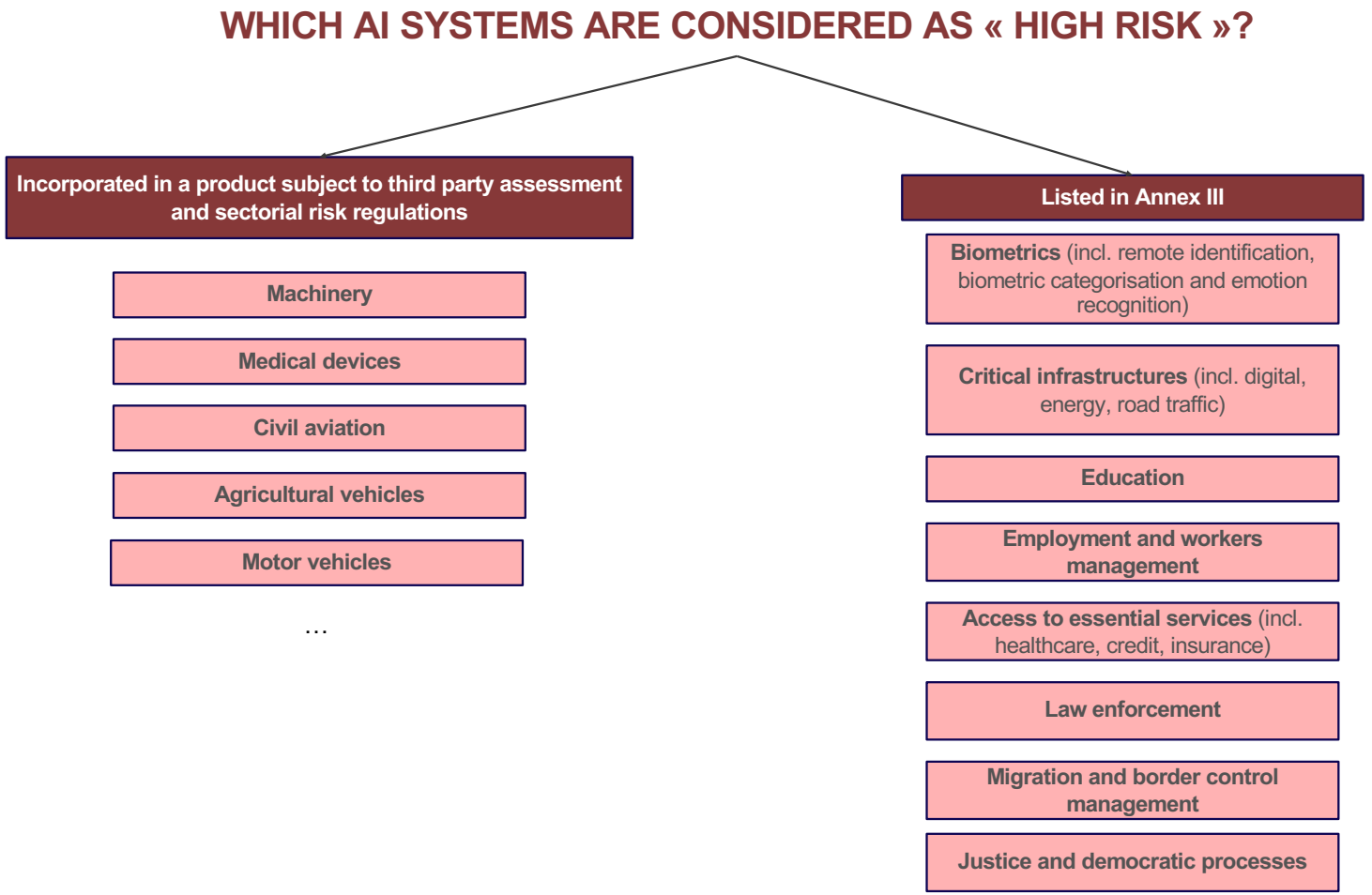
Voluntary agreement to ethics principles:

- *human agency and oversight*
- *technical robustness and safety*
- *privacy and data governance*
- *Transparency*
- *diversity, non-discrimination and fairness*
- *societal and environmental well-being*
- *accountability*

« Prohibited » AI systems

- Use of **subliminal techniques beyond a person's consciousness**
- **Exploitation of vulnerabilities of a person or group of persons** (disability, age, social situation…)
- **Social scoring**
- **Predicting** the likelihood of a natural person committing a **criminal offence**
- **Expanding facial recognition databases** thourgh untargeted scraping
- **Inferring emotions in the workplace or in education**
- **Biometric categorisation** to deduce sensitive data (race, political opinions…)
- **'Real time' remote biometric identification in public spaces for law enforcement** (except abduction, human trafficking, finding missing persons, terrorist attacks, finding suspects of criminal offences)

« High risk » AI systems (1)



« High risk » AI systems (2)

Implementation of a risk management system

Article 9

Procedures for risk assessment & mitigation measures.

Implementation of a quality management system

Article 17

Policies and procedures for regulatory compliance.

Data governance

Article 10

Providers shall put in place measures to guarantee the **quality of training, validation and testing data**:

- Relevant design choices;
- Data collection;
- Pre-processing operations (cleaning, labelling, enrichment…) ;
- Assessment and mitigation of potential biases and representativity issues ;
- Fundamental rights impact assessment ;
- Data free of errors and complete.

Exhaustive **technical documentation**

Article 11

Exhaustive documentation (content provided for in Annex IV) of the system, the conception, testing and validation processes, datasets, description of the provider's risk management system…

Record-keeping and logs conservation

Article 12

Transparency and provision of information to deployers

Article 13

Instructions for use, including metrics, capabilities to provide explanations regarding the outputs, information on the training and validation phases, risks…

Human oversight

Article 14

Effective oversight by natural persons, through human-machine interfaces, « stop » button, capacity to override the system's decision…

Precision, **robustness** and **cybersecurity** requirements

Article 15

Resilience to adversarial attacks, feedback loops, model poisoning… Assessment of metrics according to a standardised methodology.

« Certain » AI systems: transparency

"AI systems intended to interact directly with natural persons"	→	Natural persons informed that they are interacting with AI
"AI systems generating synthetic audio, image, video or text content"	→	Watermarked contents (machine readable format)
"Emotion recognition system or biometric categorisation system"	→	Concerned persons informed of the operation of the system and GDPR compliance
"AI system generating deep fake"	→	Disclose that the content has been artificially generated or manipulated (do not apply to artistic or satirical works)
"AI system generating text with the purpose of informing the public"	→	Disclose that the content has been artificially generated or manipulated

Specific rules for general purpose AI (GPAI) models and systems

- What is a GPAI model?

‘**general-purpose AI model**’ *means an AI model, including where such an AI model is* ***trained with a large amount of data using self-supervision*** *at scale, that displays* ***significant generality*** *and is capable of competently* ***performing a wide range of distinct tasks*** *regardless of the way the model is placed on the market and that* ***can be integrated into a variety of downstream systems or applications****, except AI models that are used for research, development or prototyping activities before they are released on the market*

Specific rules for all general purpose AI (GPAI) models and systems

Technical documentation

(training and testing process)

Provide information and documentation to the deployers

(capabilities and limitations of the GPAI)

Policy to comply with copyright law

(and comply with “opt-out” right of authors)

Detailed summary about the content used for training

(publicly available)

Do not apply to open source and open weight GPAI models

GPAI with « systemic risks »

« GPAI » meeting one of the two following criteria:

“High impact capabilities”

(technical tools and methodologies to be refined by delegated acts)

Presumption of “high impact capabilities”, when the cumulative amount of computation used for its training measured in FLOPs is greater than 10^25.

”Designated as such by a decision of the Commission”

Unilateral decision of the EC or following a qualified alert from the scientific panel

> ‘**systemic risk**’ *means* ***a risk*** *that is* ***specific to the high-impact capabilities of general-purpose AI models***, *having a* ***significant impact*** *on the Union market* ***due to their reach***, *or* ***due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society*** *as a whole,* ***that can be propagated*** *at scale across the value chain.*

Additional obligations for GPAI with « systemic risks »

Model evaluation and adversarial testing (standardized protocols)

Systemic risks assessment and mitigation

Tracking and reporting serious incidents to the AI office

Cybersecurity of the model and of the physical infrastructure

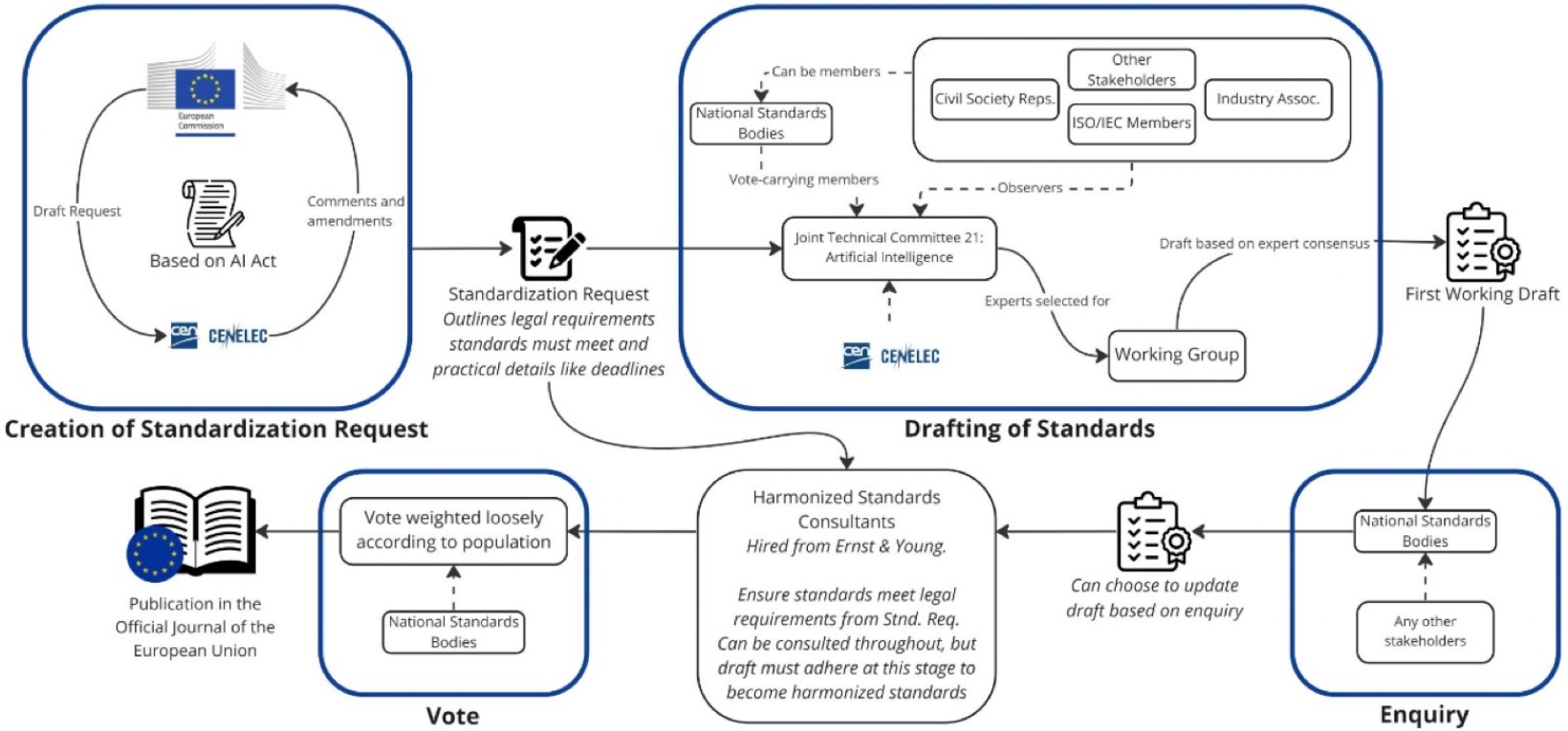
Enforcement

- **Implementation** by
  - The « **European AI office** »: expertise and capabilities, enforcement/sanctions, monitor GPAI, develop model terms of contract/encourage standards and codes of practice ;
  - The « **European AI Board** »: collaboration between member states and market surveillance authorities/notifying authorities.

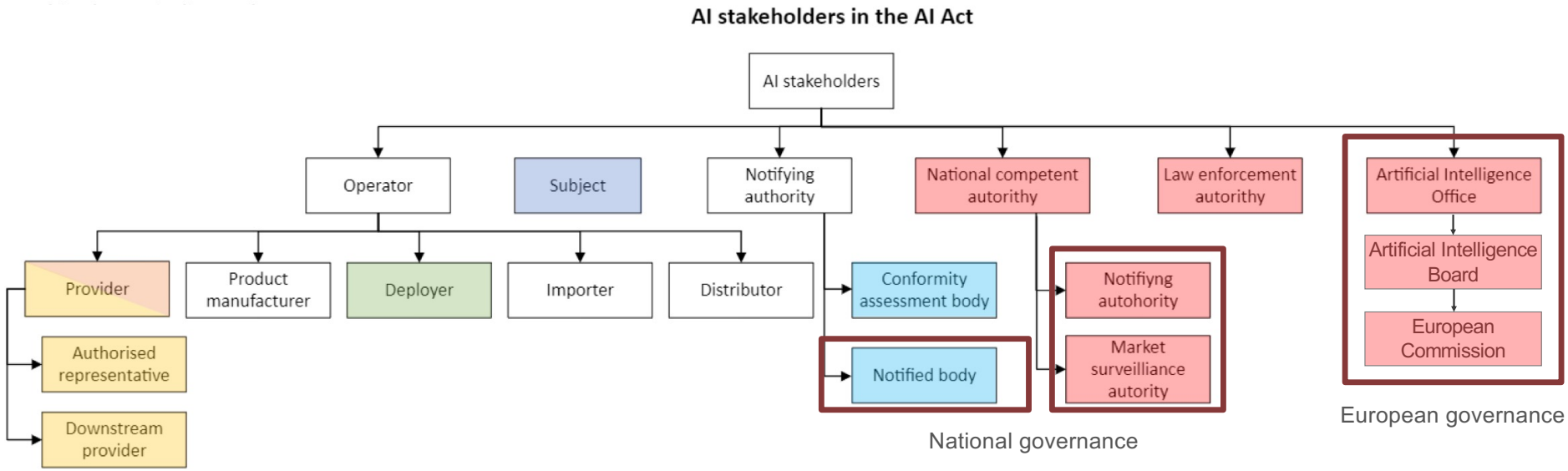
- **Regulatory sandboxes**:
  - Controlled and agreed environment for testing, including real life testing ;
  - No fines during the sandbox as long as the terms of the sandbox haven't been breached

- **Infringement and sanctions:**
  - ***Prohibited AI systems:*** up to ***35 000 000 €*** or up to ***7*** % of its total worldwide annual turnover, whichever is higher.
  - ***High risk AI systems or transparency for certain AI systems***: up to ***15 000 000*** € or up to ***3 %*** of the total worldwide annual turnover, whichever is higher***.***
  - ***No cooperation with national authorities:*** up to ***7 500 000*** € or up to ***1*** % of the total worldwide annual turnover, whichever is higher.

What's coming? Standards!



Governance


AI stakeholders in the AI Act
AI stakeholders
Operator
Subject
Notifying authority
National competent autorithy
Law enforcement autorithy
Artificial Intelligence Office
Artificial Intelligence Board
European Commission
Provider
Product manufacturer
Deployer
Importer
Distributor
Conformity assessment body
Notifiyng autohority
Authorised representative
Notified body
Market surveilliance autority
Downstream provider
National governance
European governance

Timeline



2025

2026

2027

In the shoes of a risk management team

In the shoes of a risk management team

Make groups

WHAT YOU NEED TO DO:

1. Build a short « **risk assessment** »: identify risks for individuals, physical or mental health, personal data, cyberattacks… (minimum 3), describe them and rate them based on their impact (gravity, 0 to 4) and probability of occurrence (0 to 4).

2. Think of **mitigation measures** for each risk.

Group 1: GPAI

Does **ChatGPT** pose a systemic risk?

OpenAI

ChatGPT would be considered as a GPAI system under the AI Act. Then, provider of ChatGPT-powered product or app would need to comply with the AI Act documentation's obligations.

Regarding the risks you identified, would you consider that ChatGPT poses a systemic risk?

Group 2: High-risk AI

Using AI to build an **automated HR recruitment process**

GROUPE Casino

Casino Group wants to develop an AI-driven tool that can automatically process CV, evaluate candidates et filter job applications.

This system falls within the scope of « High risk AI systems » of the AI Act.

What would be « appropriate human control measures » in this context?

Group 3: High-risk AI

Using AI to establish a (financial) **credit score**

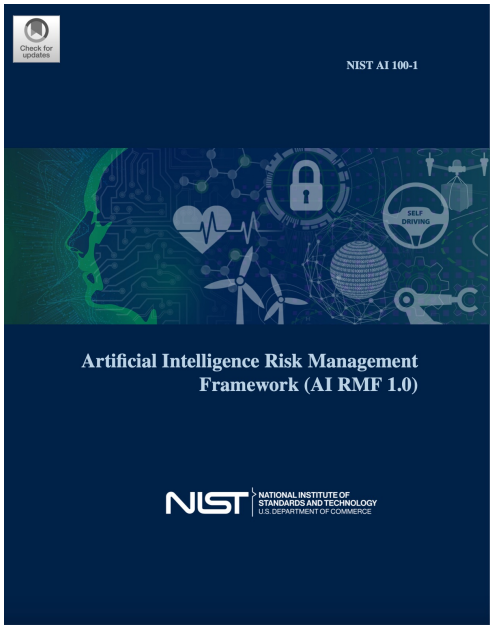
SOCIETE GENERALE

Société Générale wants to use AI to determine the creditworthiness of its customers before granting them a loan. The score is then communicated to a customer advisor who decide if he grants the loan or not.

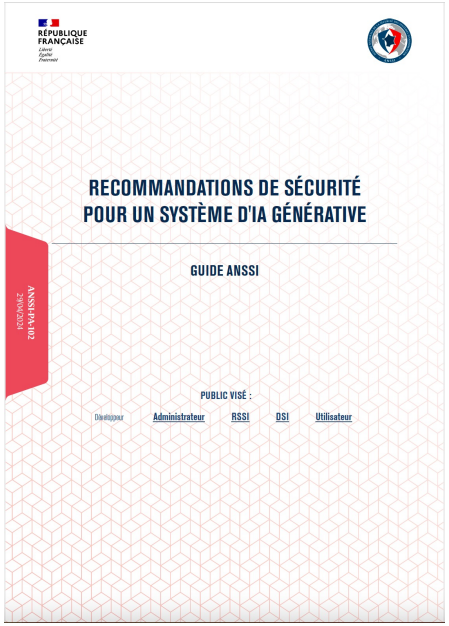
This system falls within the scope of « High risk AI systems » of the AI Act.

How would you prevent discrimination issues in this context?

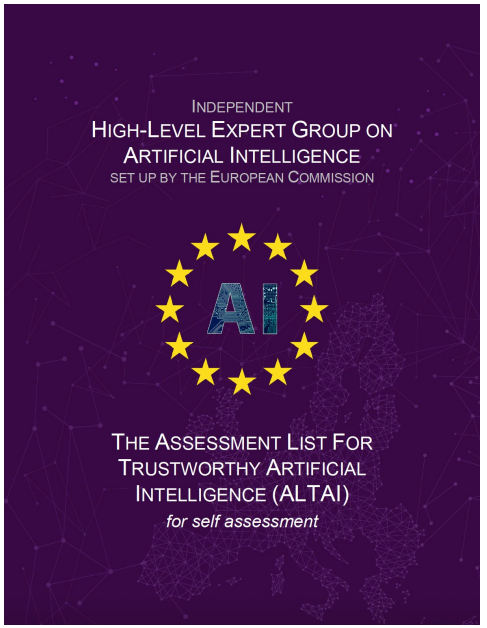
Tools to inform AI risk assessment



NIST AI RMF, 2023



ANSSI, Recommandations de sécurité pour l'IA generative, April 2024



ALTAI, HLEG on AI, 2020 (AI ethics)



